

Questions on Conditional Probability and Bayes Theorem

1. A person has undertaken a construction job. The probabilities are 0.65 that there will be a strike, 0.80 that the construction job will be completed on time if there is no strike, and 0.32 that the construction job will be completed on time if there is a strike. Determine the probability that the construction job will be completed on time.
2. Suppose 3 companies x,y,z produce TVs. x produces twice as many as y, while y and z produce same number. It is known that 2% of x, 2% of y, 4% of z are defected. All the TVs produced are put into 1 shop and then 1 TV is chosen at random. What is the probability that the TV is defected. Suppose a TV chosen is defective, what is the probability that this TV is produced by company x.
3. There are 3 boxes, the first one containing 1 white, 2 red and 3 black balls, the 2nd one containing 2W, 3R & 1B ball and the 3rd one containing 3W, 1R and 2B balls. A box is chosen at random and from it 2 balls are drawn at random. One ball is red and the other white. What is the probability that they come from the second box?
4. One percent of the population suffers from a certain disease. There is blood test for this disease, and it is 99% accurate, in other words, the probability that gives the correct answer is 0.99, regardless of whether the person is sick or healthy. A person takes the blood test, and the result says that he has the disease. The probability that he actually has the disease, is?

Binomial Distribution Questions

1. Disks produced by a company will be defective with probability 0.01. The company sells the disks in a packet of 10 offers money back guarantee if more than one defective disk found in a packet. What is the proportion of packets returned? If someone buys 3 packets, what is the probability that exactly one of them should be replaced?
2. Numbers are selected at random one at a time from the 2-digit numbers 00, 01, 02, ..., 99, with replacement. An event A occurs if and only if the product of the 2 digits of a selected number is 18. If 4 numbers are selected, find the probability that the event A occurs at least 3 times.
3. Suppose we roll a fair six-sided die 3 times. What is the probability of getting exactly 2 sixes?

Probability

Conditional probability problem

1) A die is tossed, if the number is odd on the face, what is the probability that it is prime.

Solⁿ:- Sample space, $S = \{1, 2, 3, 4, 5, 6\}$

Let E be the event that denotes getting an odd number when a die is tossed. Then

$$E = \{1, 3, 5\} \Rightarrow P(E) = \frac{3}{6} = \frac{1}{2}$$

F be the event that denotes getting a prime number when a die is tossed. Then

$$F = \{2, 3, 5\} \Rightarrow P(F) = \frac{3}{6} = \frac{1}{2}$$

$$E \cap F = \{3, 5\} \Rightarrow P(E \cap F) = \frac{2}{6} = \frac{1}{3}$$

$$P(F|E) = \frac{P(E \cap F)}{P(E)} = \frac{\frac{1}{3}}{\frac{1}{2}} = \frac{2}{3}$$

2. If A and B are two independent events of S such that $P(A) = \frac{1}{3}$, $P(B) = \frac{1}{4}$, $P(A \cup B) = \frac{1}{2}$, then find
i) $P(A|B)$ ii) $P(B|A)$

Solⁿ:- Since A and B are independent events,
 $P(A \cap B) = P(A) \cdot P(B)$
 $= \frac{1}{3} \times \frac{1}{4} = \frac{1}{12}$

$$\text{i) } P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{\frac{1}{12}}{\frac{1}{4}} = \frac{4}{12} = \frac{1}{3}$$

$$\text{ii) } P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{\frac{1}{12}}{\frac{1}{3}} = \frac{3}{12} = \frac{1}{4}$$

Total probability theorem, Baye's theorem

1) A person has undertaken a construction job. The probabilities are 0.65 that there will be a strike, 0.80 that the construction job will be completed on time if there is no strike, and 0.32 that the construction job will be completed on time if there is a strike. Determine the probability that the construction job will be completed on time.

Solⁿ:- let A be the event that the construction job will be completed on time. B be the event that there will be a strike.

We have to find $P(A)$

$$P(B) = P(\text{strike}) = 0.65$$

$$\Rightarrow P(B') = 1 - P(B) = 1 - 0.65 = 0.35$$

$$P(A|B) = 0.32 \quad P(A|B') = 0.80$$

Here B and B' form a partition of the sample space S.

Therefore by theorem on total probability

$$\begin{aligned} P(A) &= P(B) P(A|B) + P(B') P(A|B') \\ &= 0.65 \times 0.32 + 0.35 \times 0.80 \\ &= 0.488 \end{aligned}$$

2) Suppose 3 companies x, y, z produce TVs. x produces twice as many as y , while y and z produce same number. It is known that 2% of x , 2% of y , 4% of z are defected. All the TVs are produced are put into 1 shop and then 1 TV is chosen at random. What is the probability that the TV is defected. Suppose a TV chosen is defective, what is the probability that this TV is produced by company x .

Solⁿ:- let X denote the event
tv is produced by company x
let Y denote the event
tv is produced by company y
let Z denote the event
tv is produced by company z .
let D denote the event
tv is defected.

Given $P(X) = 0.5$, $P(Y) = 0.25$, $P(Z) = 0.25$

$$P(D|X) = 0.02$$

$$P(D|Y) = 0.02$$

$$P(D|Z) = 0.04$$

$$\begin{aligned} \text{i) } P(D) &= P(X) P(D|X) + P(Y) P(D|Y) \\ &\quad + P(Z) P(D|Z) \end{aligned}$$

$$\begin{aligned} &= 0.5(0.02) + 0.25(0.02) + 0.25(0.04) \\ &= 0.025 \end{aligned}$$

The probability that the tv is detected is 0.025

$$\begin{aligned} \text{ii) } P(X|D) &= \frac{P(X) P(D|X)}{P(X) P(D|X) + P(Y) P(D|Y) \\ &\quad + P(Z) P(D|Z)} \end{aligned}$$

$$\begin{aligned} &= \frac{0.5(0.02)}{0.5(0.02) + 0.25(0.02) + 0.25(0.04)} \\ &= 0.4 \end{aligned}$$

3) There are 3 boxes, the first one containing 1 white, 2 red and 3 black balls, the 2nd one containing 2W, 3R & 1 B ball and the 3rd one containing 3W, 1R and 2B balls. A box is chosen at random. and from it 2 balls are drawn at random. One ball is red and the other white. What is the probability that they come from the second box?

Solⁿ:- Let B_1 be the event of choosing ball from box 1

B_2 be the event of choosing ball from box 2

B_3 be the event of choosing ball from box 3.

$$P(B_1) = \frac{1}{3}, \quad P(B_2) = \frac{1}{3}, \quad P(B_3) = \frac{1}{3}$$

Let A be the event that one red ball and one white ball is drawn.

We have to find $P(B_2|A)$

Total no of ways to choose two balls from six balls in any box is

$$\binom{6}{2} = 15$$

For box 1 (1W, 2R, 3B):-

The no of ways to choose one white and one red ball is

$$\binom{1}{1} \times \binom{2}{1} = 2$$

$$\therefore P(A|B_1) = \frac{2}{15}$$

For box 2 (2W, 3R, 1B):-

The no of ways to choose one white and one red ball is

$$\binom{2}{1} \times \binom{3}{1} = 6$$

$$\therefore P(A|B_2) = \frac{6}{15} = \frac{2}{5}$$

For box 3 (2W, 3R, 1B):-

The no of ways to choose one white and one red ball is

$$\binom{2}{1} \times \binom{3}{1} = 6$$

$$\therefore P(A|B_3) = \frac{6}{15} = \frac{2}{5}$$

$$P(A) = P(A|B_1) P(B_1) + P(A|B_2) P(B_2) + P(A|B_3) P(B_3)$$

$$P(A) = \frac{2}{15} \times \frac{1}{3} + \frac{2}{5} \times \frac{1}{3} + \frac{2}{5} \times \frac{1}{3}$$

$$= \frac{11}{45}$$

By Baye's theorem

$$P(B_2|A) = \frac{P(A|B_2) \cdot P(B_2)}{P(A)}$$

$$= \frac{\left(\frac{6}{15}\right) \left(\frac{1}{3}\right)}{\left(\frac{11}{45}\right)} = \frac{6}{11}$$

4) One percent of the population suffers from a certain disease. There is blood test for this disease, and it is 99% accurate, in other words, the probability that gives the correct answer is 0.99, regardless of whether the person is sick or healthy. A person takes the blood test, and the result says that he has the disease. The probability that he actually has the disease, is?

A be the event of having the disease
B be the event of testing +ve.

$$P(A) = 0.01$$

$$\begin{aligned} P(B) &= P(B|A)P(A) + P(B|A')P(A') \\ &= 0.01 \times 0.99 + 0.01 \times 0.99 \\ &= 0.0198 \end{aligned}$$

$$P(A \cap B) = 0.99 \times 0.01$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{0.99 \times 0.01}{0.0198} = \underline{\underline{0.5}}$$

Random variable

Ex:- Consider the random experiment of tossing a die.

$$S = \{1, 2, 3, 4, 5, 6\}$$

Let $X = S \rightarrow R$ be a function defined by $X(S) = 10S$

$$X(1) = 10, \quad X(2) = 20, \quad \dots, \quad X(6) = 60$$

X is a Random variable.

Suppose $Y = S \rightarrow R$ is a function defined by

$$X(S) = 0 \quad \text{if } S \text{ is odd}$$

$$X(S) = 1 \quad \text{if } S \text{ is even.}$$

Then X takes value either 0 or 1.

$\therefore X$ is a Random variable.

Bernoulli's distribution.

1) A machine produces items and each item is defective with probability 0.1 and non-defective with probability 0.9

a) Define Bernoulli's r.v X for item is defective

b) Find pmf, expectation and variance.

Solⁿ:- a) Bernoulli's random variable

$$X(\text{defective item}) = 1$$

$$X(\text{non-defective}) = 0$$

$$b) \text{ pmf} = p(x) = (0.1)^x (0.9)^{1-x} \quad x=0,1$$

$$E[X] = p = 0.1$$

$$\begin{aligned} \text{Var}(X) &= p(1-p) = 0.1(1-0.1) = 0.1 \times 0.9 \\ &= 0.09 \end{aligned}$$

2) A student guesses the answer to a true or false question.

a) Define Bernoulli's r.v X for answer is correct.

b) Find pmf, expectation and variance.

Solⁿ: Random variable is

$$\begin{aligned} \text{a) } X(\text{correct answer}) &= 1 \\ X(\text{wrong answer}) &= 0 \end{aligned}$$

b) Here the probability that the answer is correct is 0.5

$$P(X=1) = 0.5$$

$$P(X=0) = 1 - 0.5 = 0.5$$

$$\text{pmf} = P(X) = (0.5)^x (0.5)^{1-x} \quad x=0,1$$

$$E(X) = p = 0.5$$

$$\text{Var}(X) = p(1-p) = 0.5 \times 0.5 = 0.25$$

Binomial distribution

1) Disks produced by a company will be defective with probability 0.01. The company sells the disks in a packet of 10 offers money back guarantee if more than one defective disk found in a packet. What is the proportion of packets returned? If someone buys 3 packets, what is the probability that exactly one of them should be replaced?

Given:-

- Probability a disk is defective = $p = 0.01$
- Number of disks per packet = $n = 10$
- A packet is returned if more than one defective disk is found.

∴ X is binomial ($n = 10, p = 0.01$)

We have to find

$$\begin{aligned} P(\text{Packet returned}) &= P(X > 1) \\ &= 1 - P(X \leq 1) \\ &= 1 - [P(X = 0) + P(X = 1)] \end{aligned}$$

$$P(X = x) = \binom{n}{x} p^x q^{n-x}, \quad x = 0, 1, 2, \dots, n$$

$$\begin{aligned} P(X = 0) &= \binom{10}{0} (0.01)^0 (0.99)^{10} = (0.99)^{10} \\ &= 0.90438 \end{aligned}$$

$$P(X = 1) = \binom{10}{1} (0.01)^1 (0.99)^9 = 0.09135$$

$$\begin{aligned} \therefore P(X \leq 1) &= 0.90438 + 0.09135 \\ &= 0.99573 \end{aligned}$$

$$\therefore P(\text{Packet returned}) = 0.00427$$

(b) Probability that exactly one of three packets should be replaced.

Let the event "packet returned" = success
 $= p_n = 0.00427$

Now Y is binomial ($n=3$, $p=0.00427$)

We want $P(Y=1) = \binom{3}{1} p_n (1-p_n)^2$

$$P(Y=1) = 3 (0.00427) (1-0.00427)^2 \\ = 0.0127$$

2) Numbers are selected at random one at a time from the 2-digit numbers 00, 01, 02, ..., 99, with replacement. An event A occurs if and only if the product of the 2 digits of a selected number is 18. If 4 numbers are selected, find the probability that the event A occurs at least 3 times.

Solⁿ: Digits are from 0 to 9. We need ordered digit pairs (a, b) with $a \times b = 18$. The valid ordered pairs are $(2, 9), (9, 2), (3, 6), (6, 3)$. Therefore 4 favourable digits out of 100.

$$\text{So } p = P(A) = \frac{4}{100} = \frac{1}{25}$$

$$\therefore X \sim \text{Binomial} (n=4, p = \frac{1}{25})$$

We want to find $P(X \geq 3)$

$$\begin{aligned} \therefore P(X \geq 3) &= P(X=3) + P(X=4) \\ &= \binom{4}{3} p^3 (1-p)^{4-3} + \binom{4}{4} p^4 (1-p)^{4-4} \\ &= 4 p^3 (1-p) + p^4 \\ &= 4 \left(\frac{1}{25} \right)^3 \left(1 - \frac{1}{25} \right) + \left(\frac{1}{25} \right)^4 \\ &= 4 \left(\frac{1}{25^3} \right) \left(\frac{24}{25} \right) + \frac{1}{(25)^4} \\ &= \frac{97}{(25)^4} = \frac{97}{390625} \end{aligned}$$

3. Suppose we roll a fair six-sided die 3 times. What is the probability of getting exactly 2 sixes?

Solⁿ:- Let X be no. of sixes obtained in 3 rolls.

Number of trials $= n = 3$

Probability of success (getting a six) $= p = \frac{1}{6}$

Then $X \sim \text{Binomial}(n=3, p=1/6)$

We have to find $P(X=3)$

$$\begin{aligned} P(X=3) &= \binom{3}{3} \left(\frac{1}{6}\right)^3 \left(1 - \frac{1}{6}\right)^{3-3} \\ &= \frac{1}{6^3} \left(\frac{5}{6}\right)^0 = \frac{1}{6^3} = \frac{1}{216} \end{aligned}$$

Normal distribution

Probability density function

$$f(x) = \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{1}{2} \left(\frac{x-\mu}{\sigma}\right)^2}, \quad -\infty < x < \infty$$

$\sigma \rightarrow$ standard deviation

$\mu \rightarrow$ mean

Properties

1) If $X_1, X_2, \dots, X_n$ are independent R.V that are normally distributed, then sum S is also normally distributed.

$$S = X_1 + X_2 + \dots + X_n$$

$$E(S) = E(X_1) + E(X_2) + \dots + E(X_n)$$

$$V(S) = V(X_1) + V(X_2) + \dots + V(X_n)$$

2) If $X_1, X_2, \dots, X_n$ are independent R.V that are normally distributed, then the R.V. Q is defined as $Q = a_1 X_1 + a_2 X_2 + \dots + a_n X_n + b$ will also be normally distributed with

$$E(Q) = a_1 E(x_1) + a_2 E(x_2) + \dots + a_n E(x_n) + b$$

$$V(Q) = a_1^2 V(x_1) + a_2^2 V(x_2) + \dots + a_n^2 V(x_n)$$

Problems:-

1. The weight of a module used in a spacecraft is to be closely controlled. Since the module uses a bolt-nut-washer assembling in numerous places, a study was conducted to find the distribution of weights of these parts. It was found that the three weights, in grams, are normally distributed with the following means and variance.

	Means	Variances
Bolt	312.8	2.67
Nut	53.2	0.85
Washer	17.5	0.21

Find the distribution of the weight of the assembly. Find mean, variance and standard deviation.

Solⁿ:- The weight of the assembly is the sum of the weights of the 3 component parts, which are 3 random variables.

The individual weights are independent since the weight of any one component part does not influence the weight of the other two.

$\therefore$ The weight of the assembly follow a normal distribution.

mean weight of the assembly

$$= 312.8 + 53.2 + 17.5 = 383.5 \text{ grams}$$

$$\text{Variance} = 2.67 + 0.85 + 0.21 = 3.73 \text{ gm}^2$$

$$\text{S.D} = \sqrt{\text{Variance}} = \sqrt{3.73} = 1.93 \text{ gms}$$

2. The four independent normal r.v X_1, X_2, X_3 and X_4 have the following means and variances.

	Mean	Variance
X_1	12	4
X_2	-5	2
X_3	8	5
X_4	10	1

Find mean and variance of

$$Q = X_1 - 2X_2 + 3X_3 - 4X_4 + 5. \text{ Find also}$$

the S.D of Q.

Soln:-

$$\begin{aligned} \text{Mean} &= E(Q) = 1E(X_1) - 2E(X_2) + 3E(X_3) - 4E(X_4) + 5 \\ &= 1 \times 12 - 2 \times (-5) + 3 \times 8 - 4(10) + 5 \\ &= 11 \end{aligned}$$

$$\begin{aligned} \text{Variance } V(Q) &= 1^2 V(X_1) + (-2)^2 V(X_2) + 3^2 V(X_3) \\ &\quad + (-4)^2 V(X_4) \\ &= 1(4) + 4(2) + 9(5) + 16(1) \\ &= 73 \end{aligned}$$

$$\text{S.D} = \sqrt{73} = 8.544$$

